

10. Expectation Maximization Algorithm

Student Performance Clustering dataset.

```
from sklearn.mixture import GaussianMixture
```

```
X = [[2,65],[3,70],[4,75],[5,80],[6,85],  
     [7,88],[8,90],[9,93],[10,95],[11,98]]
```

```
model = GaussianMixture(n_components=2, random_state=42)
```

```
model.fit(X)
```

```
print("Cluster Labels:", model.predict(X))
```

```
print("New Sample Cluster:", model.predict([[7,89]])[0])
```

OUTPUT

```
= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB  
lab/EXP 7.py  
Cluster Labels: [0 0 0 0 0 1 1 1 1 1]  
New Sample Cluster: 1  
|
```